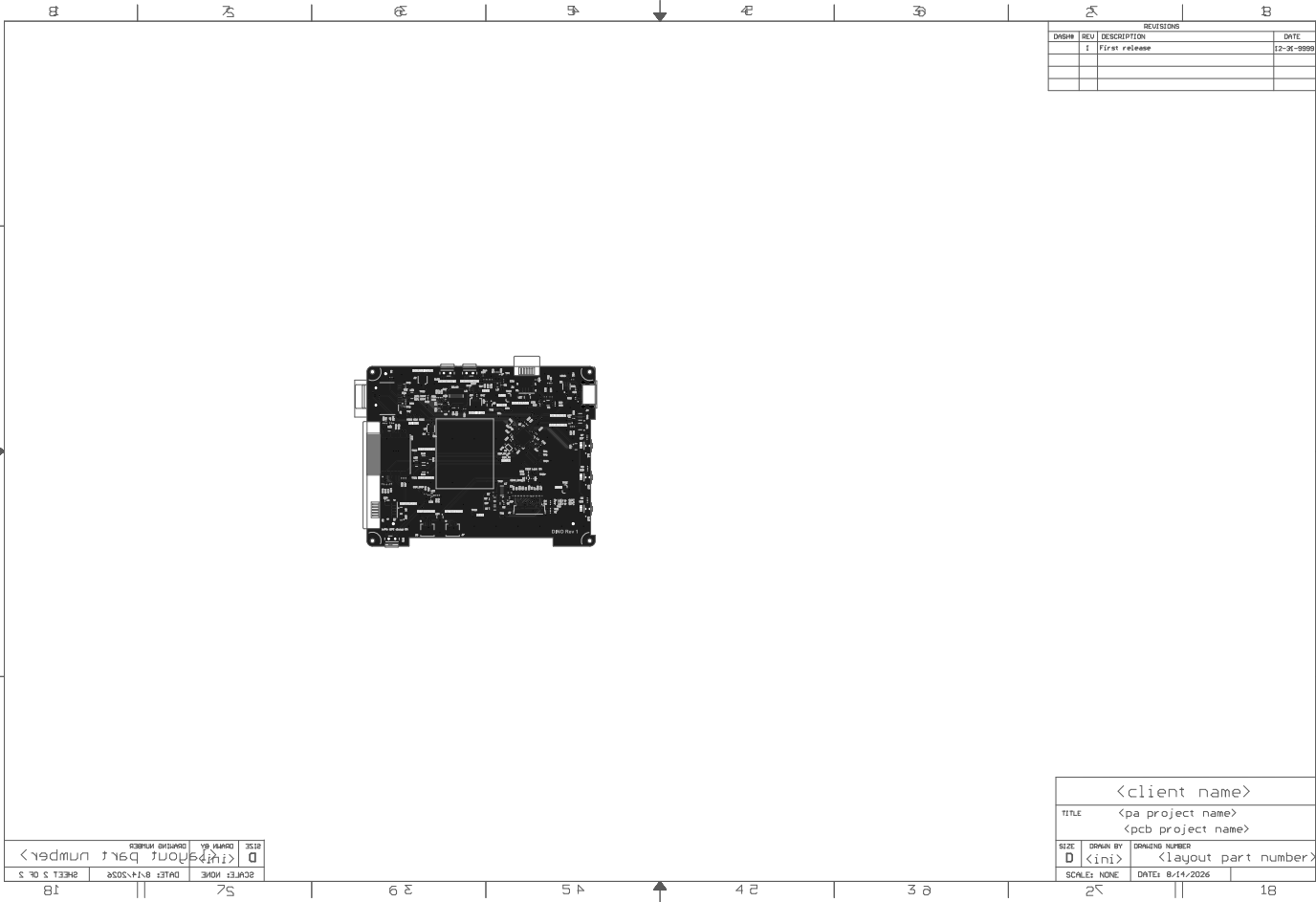
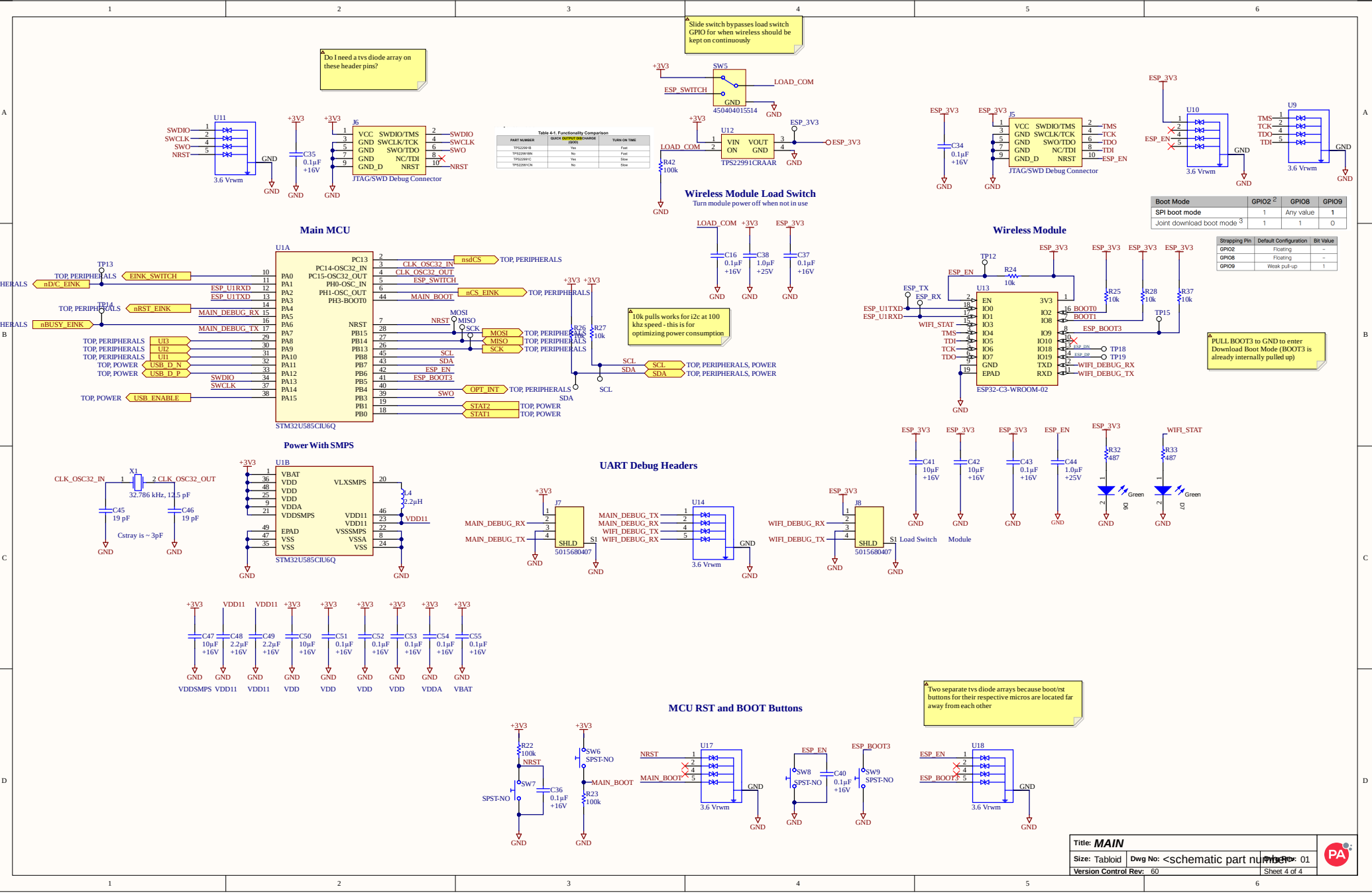


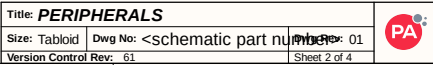


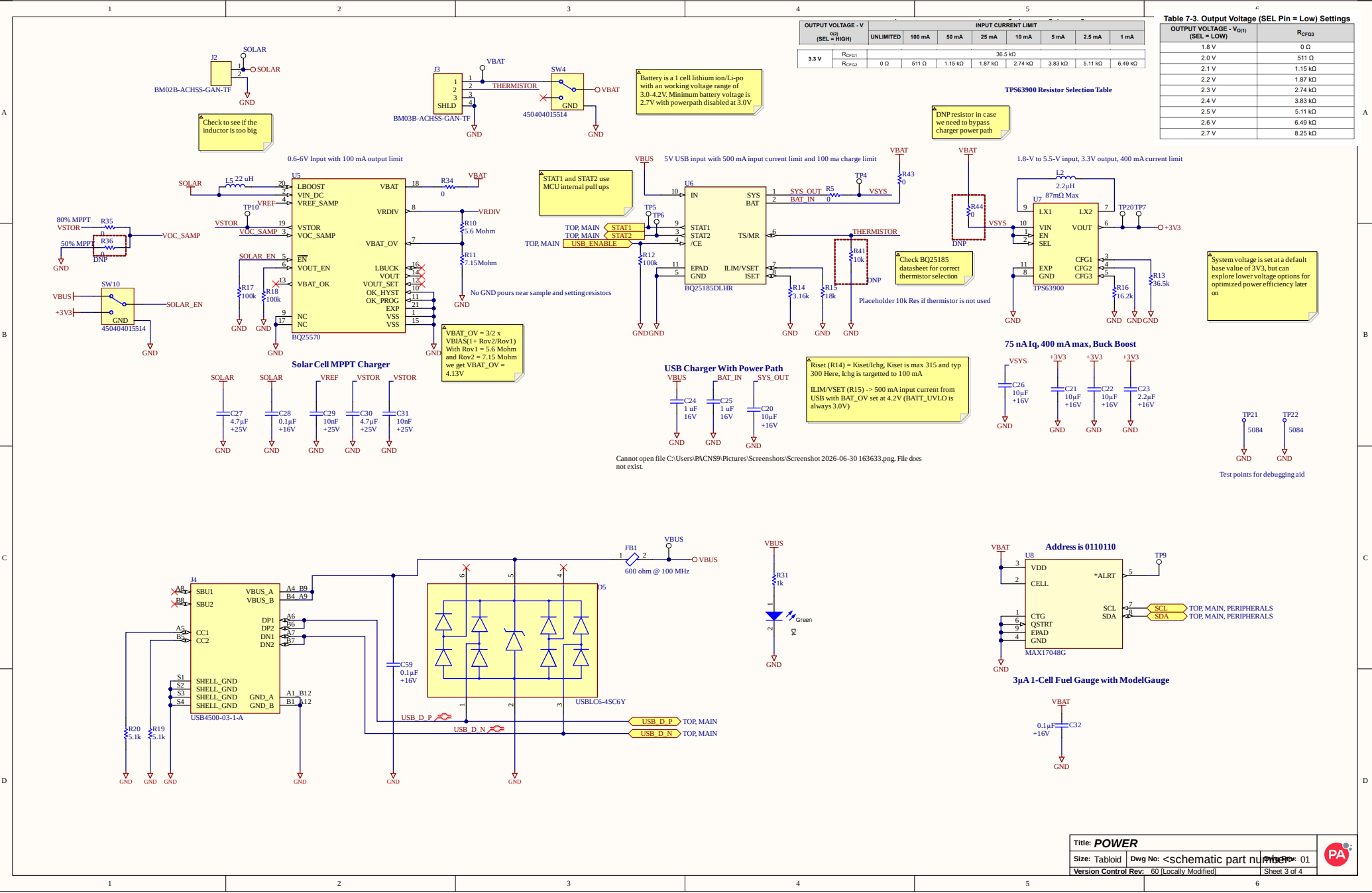
|                                               |                   |                                        |
|-----------------------------------------------|-------------------|----------------------------------------|
| <client name>                                 |                   |                                        |
| TITLE <pa project name><br><pcb project name> |                   |                                        |
| SIZE<br>D                                     | DRAWN BY<br><ini> | DRAWING NUMBER<br><layout part number> |
| SCALE: NONE                                   |                   | DATE: 8/14/2026                        |

[illegible]

22. ORIENTATION PLANE LAYERS (GPx) ARE NEGATIVE  
ALL OTHER LAYERS ARE POSITIVE







| OUTPUT VOLTAGE - V              |                   | INPUT CURRENT LIMIT |        |         |         |         |         |         |         |
|---------------------------------|-------------------|---------------------|--------|---------|---------|---------|---------|---------|---------|
| O <sub>Q2</sub><br>(SEL = HIGH) |                   | UNLIMITED           | 100 mA | 50 mA   | 25 mA   | 10 mA   | 5 mA    | 2.5 mA  | 1 mA    |
|                                 |                   |                     |        |         |         |         |         |         |         |
| 3.3 V                           | R <sub>CPQ1</sub> | 36.5 kΩ             |        |         |         |         |         |         |         |
|                                 | R <sub>CPQ2</sub> | 0 Ω                 | 511 Ω  | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |

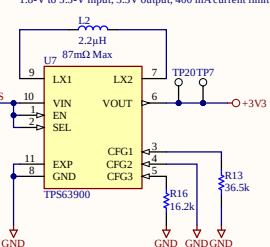
Table 7-3. Output Voltage (SEL Pin = Low) Settings

| OUTPUT VOLTAGE - V <sub>OUT</sub> (SEL = LOW) | R <sub>CP2</sub> |
|-----------------------------------------------|------------------|
| 1.8 V                                         | 0 Ω              |
| 2.0 V                                         | 511 Ω            |
| 2.1 V                                         | 1.15 kΩ          |
| 2.2 V                                         | 1.87 kΩ          |
| 2.3 V                                         | 2.74 kΩ          |
| 2.4 V                                         | 3.83 kΩ          |
| 2.5 V                                         | 5.11 kΩ          |
| 2.6 V                                         | 6.49 kΩ          |
| 2.7 V                                         | 8.25 kΩ          |

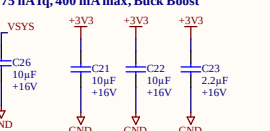
TPS63900 Resistor Selection Table

| OUTPUT VOLTAGE - V | R <sub>CP2</sub> |
|--------------------|------------------|
| 1.8 V              | 0 Ω              |
| 2.0 V              | 511 Ω            |
| 2.1 V              | 1.15 kΩ          |
| 2.2 V              | 1.87 kΩ          |
| 2.3 V              | 2.74 kΩ          |
| 2.4 V              | 3.83 kΩ          |
| 2.5 V              | 5.11 kΩ          |
| 2.6 V              | 6.49 kΩ          |
| 2.7 V              | 8.25 kΩ          |

1.8-V to 5.5-V input, 3.3V output, 400 mA current limit



75 nA Iq, 400 mA max, Buck Boost



System voltage is set at a default base value of 3V3, but can explore lower voltage options for optimized power efficiency later on

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# Board Stack Report